FOOT & ANKLE · NEET-SS / FET / INI-SS

Achilles and tendons

Part A: Achilles tendon

1. Anatomy

- **Formed by the gastrocnemius (medial and lateral heads: cross the knee) and soleus** (± plantaris medially); **largest and strongest tendon in the body** (loads up to **about 12 times body weight** in running).
- **Fibres rotate about 90 degrees** (medial fibres become posterior, **soleus fibres insert medially**) as they descend: stress concentration.
- **Paratenon** (no true synovial sheath), **insertion on the middle third of the posterior calcaneal tuberosity** (footprint); **retrocalcaneal bursa** (between tendon and calcaneus) and **retro-Achilles (subcutaneous) bursa**.
- **Blood supply: posterior tibial artery (proximal and distal)**, **peroneal artery (mid-portion)**; **hypovascular watershed zone about 2-6 cm proximal to the insertion** (**site of most ruptures and mid-portion tendinopathy**).
- **Sural nerve**: crosses from medial to lateral, **lateral to the tendon at about 10 cm proximal to the insertion** (**at risk in lateral incisions and percutaneous repairs**).

2. Acute Achilles tendon rupture

- **Epidemiology: men aged 30-50 ("weekend warriors")**, **sudden eccentric loading** (push-off with knee extended, unexpected dorsiflexion), **left more than right**; **2-6 cm above the insertion (commonest)**.

- **Risk factors: fluoroquinolones**, **corticosteroids (systemic or local injection)**, **pre-existing tendinopathy**, inflammatory arthritis, gout, **renal failure, hyperparathyroidism**, previous contralateral rupture.
- **Clinical: "kicked in the back of the leg"**, **audible pop**, **weak plantarflexion (but often preserved through FHL, FDL, TP, peronei)**: **missed in about 20%**.
  - **Simmonds-Thompson test (calf squeeze: absent plantarflexion = rupture)** (**most reliable**),
  - **Palpable gap**,
  - **Matles test** (**loss of resting plantarflexion tension: the injured foot hangs in neutral/dorsiflexion with knees flexed prone**),
  - **Increased passive dorsiflexion**, **O'Brien needle test**, **Copeland sphygmomanometer test**.
  - **Two of the clinical tests (AAOS guideline) sufficient for diagnosis.**
- **Imaging: ultrasound** (gap, **gap distance in plantarflexion (over 5-10 mm)**: may guide treatment); **MRI** (chronic/equivocal); **radiograph: Kager's fat pad distortion (triangle obscured), avulsion fracture (calcaneal tuberosity avulsion: "beak" fracture: urgent surgery: skin necrosis)**.

Management

	Non-operative (functional rehabilitation)	Operative (open or minimally invasive)
Re-rupture	**About 3-12% (similar to surgery when early functional rehab is used)**	**About 2-5%**

	Non-operative (functional rehabilitation)	Operative (open or minimally invasive)
Complications	Re-rupture; **tendon elongation (weaker push-off)**	**Wound complications/infection (about 5-10% open), sural nerve injury (percutaneous: about 5-10%)**, DVT
Strength	Possibly slightly weaker (elongation)	Slightly better plantarflexion strength in some studies

- **Evidence: early functional rehabilitation (early weight bearing in an equinus boot with heel wedges)** reduces the re-rupture difference: **Willits (2010)**, **UKSTAR trial (2020, UK: functional bracing vs plaster casting: similar outcomes)**, **Scandinavian trials (Nilsson-Helander, Ochen meta-analysis)**: **surgery reduces re-rupture absolute risk by about 2-3% at the cost of more complications**; **choice individualised** (**athletes, delayed presentation, large gap, avulsion: surgery**).
- **Non-operative protocol: equinus (about 20-30 degrees plantarflexion) in a boot with wedges**, **early protected weight bearing (2 weeks)**, **gradual reduction of equinus over 6-8 weeks**, **avoid passive dorsiflexion stretching for 12 weeks**; return to sport about 6-9 months.
- **Operative: open repair (Krackow/Bunnell core sutures + epitendinous; medial incision avoids the sural nerve)**, **minimally invasive/percutaneous (Achillon, PARS, Ma-Griffith)** (**fewer wound problems, sural nerve risk**); **repair with the ankle in resting tension (compare with the contralateral side)**.
- **Calcaneal tuberosity (sleeve) avulsion fracture: urgent ORIF (skin at risk)**.

3. Chronic (neglected) Achilles rupture

- **Over 4-6 weeks**; **gap with scar**; **weak push-off, inability to single heel rise**, **increased passive dorsiflexion**.
- **Kuwada classification** (gap): **I partial**, **II complete with gap under 3 cm (end-to-end repair)**, **III gap 3-6 cm (V-Y advancement / gastrocnemius recession / tendon graft)**, **IV gap over 6 cm (tendon transfer + graft)**.
- **Myerson: under 1-2 cm: end-to-end repair**; **2-5 cm: V-Y lengthening (gastrocnemius aponeurosis) ± FHL transfer**; **over 5 cm: FHL transfer (± turndown flap, allograft)**.
- **FHL transfer** (**strongest, in phase, adjacent, vascularised muscle belly brings blood supply, low donor morbidity**) to the calcaneus (**tunnel**); **FDL, peroneus brevis** alternatives; **gastrocnemius turndown flap (Bosworth/Lindholm)**.

4. Achilles tendinopathy

	Mid-portion (non-insertional)	Insertional
Location	**2-6 cm proximal to the insertion**	**At the calcaneal insertion**
Patients	**Runners, active middle-aged**	**Older, less active, overweight**, inflammatory arthritis (seronegative)
Pathology	**Tendinosis (degeneration, neovascularisation)**, **paratenonitis** (crepitus, fixed tender area)	**Calcification/enthesophyte (spur)**, **Haglund deformity (posterosuperior calcaneal prominence)**, **retrocalcaneal bursitis**
Signs	**Painful fusiform swelling that moves with the tendon (painful arc sign)**; **Royal London Hospital test**	**Posterior heel tenderness, bump ("pump bump")**

	Mid-portion (non-insertional)	Insertional
First-line	**Eccentric loading (Alfredson protocol: 3 x 15 reps, twice daily, 12 weeks, knee straight and bent)** or **heavy slow resistance**; **ESWT**; **high-volume injection**, sclerotherapy	**Eccentric loading on flat ground (not into dorsiflexion)**, **heel lift**, **shoe modification (open-backed)**, **ESWT**
Surgery (after 6 months)	**Debridement/longitudinal tenotomies**, **stripping of paratenon/neovessels**, **plantaris release (medial)**, **gastrocnemius recession**, **FHL transfer if over 50% of tendon debrided**	**Haglund resection + debridement of calcification + reattachment (suture anchors)** (**central tendon-splitting approach**); **FHL transfer** for extensive degeneration (over 50%); **Zadek (dorsal closing-wedge calcaneal) osteotomy**

- **Haglund syndrome: Haglund deformity + retrocalcaneal bursitis + insertional tendinopathy**; **radiographic: Fowler-Philip angle (over 75 degrees)**, **parallel pitch lines (Pavlov)**.
- **Avoid steroid injections into the Achilles** (rupture); **fluoroquinolones** caution.

Part B: Other tendons

5. Peroneal tendons

- **Anatomy: peroneus brevis (anterior/medial, closer to the fibula, in the retromalleolar groove; inserts on the base of the fifth metatarsal)**, **peroneus longus (posterior/lateral; crosses under the cuboid in the cuboid tunnel to insert on the base of the first metatarsal and medial cuneiform)**; **superior peroneal retinaculum (SPR: from the posterolateral fibula to the**

calcaneus); **fibrocartilaginous rim** deepens the groove; **os peroneum** (sesamoid in PL at the cuboid: about 20%).

- **Peroneus brevis split tears: longitudinal splits from compression against the fibula** (**cavovarus foot, ankle instability, low-lying PB muscle belly, peroneus quartus**): **lateral ankle pain/swelling**; **MRI ("boomerang/chevron" shape)**; **debridement and tubularisation (if under 50% involved)**, **tenodesis to PL (if over 50%)**; **correct cavovarus**.
- **Peroneal tendon subluxation/dislocation: forceful dorsiflexion with peroneal contraction (skiing)**; **SPR avulsion with a "fleck sign" (rim fracture of the lateral malleolus: Eckert-Davis grade 3)**; **Eckert-Davis/Oden grades (1: SPR stripped from the fibula with periosteum (commonest); 2: fibrocartilaginous rim elevated with SPR; 3: bony avulsion; 4: SPR torn from calcaneal attachment)**. **Acute: cast trial (about 50% success) or SPR repair**; **chronic: SPR repair/reconstruction + fibular groove deepening**. **Intrasheath subluxation** (tendons swap position within the groove).
- **Painful os peroneum syndrome (POPS): PL rupture at the cuboid notch/os peroneum fracture**; **proximal migration of the os peroneum on serial radiographs = PL rupture**; excision + PB tenodesis.

6. Tibialis posterior

- See flatfoot notes (PCFD).

7. Flexor hallucis longus

- **FHL tenosynovitis ("dancer's tendinitis")**: **ballet dancers (en pointe)**: **posteromedial ankle pain, triggering of the hallux ("hallux saltans")**, **pain on resisted hallux flexion**; **stenosis at the fibro-osseous tunnel behind the talus (between the medial and lateral talar tubercles)**; associated with **posterior ankle impingement (os trigonum/Stieda process)**: **rest, physiotherapy;**

release of the FHL tunnel ± os trigonum excision (open posteromedial or arthroscopic/endoscopic posterior).

- **Knot of Henry**: FHL-FDL crossover (plantar midfoot).

8. Tibialis anterior

- **Rupture: elderly men, degenerative**; see flatfoot notes.

9. Posterior ankle impingement

- **Os trigonum** (**unfused lateral tubercle of the posterior talar process: about 7-14%**) or **Stieda process** (elongated lateral tubercle); **ballet dancers, footballers (forced plantarflexion: nutcracker)**; **posterior ankle pain in plantarflexion (forced plantarflexion test)**, **diagnostic injection**; **excision (posterior arthroscopy: posterolateral and posteromedial portals with the patient prone; FHL is the landmark: stay lateral to FHL to protect the tibial neurovascular bundle)**.
- **Anterior ankle impingement: "footballer's ankle"**: **anterior tibial and talar neck osteophytes (kissing)** (**bony**) or **anterolateral soft-tissue impingement (Bassett ligament: accessory anteroinferior tibiofibular ligament fascicle; meniscoid lesion (Wolin))**; **arthroscopic debridement**; **anterior arthroscopy portals: anteromedial (medial to the tibialis anterior) and anterolateral (lateral to peroneus tertius: superficial peroneal nerve (intermediate dorsal cutaneous branch) at risk: commonest complication of ankle arthroscopy)**; **avoid the anterocentral portal (dorsalis pedis/deep peroneal nerve)**.

10. Plantar fascia rupture

- **Acute rupture** (often after steroid injection), **plantar heel pain and bruising**; non-operative (boot).

11. Gastrocnemius-soleus problems

- **Tennis leg** (**medial head of gastrocnemius tear at the musculotendinous junction**; middle-aged; "shot in the calf"); **differentiate from DVT (ultrasound)**; non-operative.
- **Isolated gastrocnemius contracture** (**Silfverskiöld test**): associated with **metatarsalgia, plantar fasciitis, Achilles tendinopathy, PCFD, midfoot arthritis**: **gastrocnemius recession (Strayer: distal aponeurosis; Baumann: intramuscular proximal)** (**sural nerve at risk**); **Achilles lengthening (Hoke triple hemisection: percutaneous)** when both gastrocnemius and soleus are tight (**overlengthening: calcaneal gait/weak push-off**).

Exam pearl: **Achilles: watershed 2-6 cm above insertion; sural nerve lateral at about 10 cm. Thompson test most reliable. Early functional rehab narrows re-rupture difference (UKSTAR, Willits); surgery: fewer re-ruptures, more wound/sural complications. Chronic gap over 5 cm: FHL transfer (± V-Y/turndown).**

Exam pearl: **Mid-portion tendinopathy: eccentric Alfredson (12 weeks), plantaris release. Insertional: no dorsiflexion eccentrics; Haglund resection + debridement + reattachment; FHL if over 50% debrided. Avoid steroid injections and fluoroquinolones.**

Exam pearl: **Peroneus brevis split tears (cavovarus): under 50% repair, over 50% tenodesis to PL. Peroneal dislocation: SPR, fleck sign; groove deepening for chronic. Os peroneum migration = PL rupture. Dancers: FHL tendinitis + os trigonum (stay lateral to FHL arthroscopically). Ankle arthroscopy: anterolateral portal risks the superficial peroneal nerve.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.